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Wisecaver Lab algal DNA Extraction with Plant DNAzol Reagent

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We use this protocol and it's working

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Abstract

Here, we present a protocol to extract DNA from *Prymnesium parvum* (Haptophyta) and other eukaryotic microalgae maintained in liquid culture. The protocol utilizes a chloroform phase extraction followed by several washing steps. DNA isolated via this methods is appropriate for short-read Illumina sequencing and molecular biology applications such as PCR.

Prepare Station and Materials

- 1 Perform cell counts using the flow cytometer. Make sure you reserve time on the Accuri Teams calendar.
- 1.1 Dilute  500 μL culture in  500 μL algal medium.
- 1.2 Get initial sample weight on the analytical balance (the one with the enclosed hood)
- 1.3 Open the Accuri C6 Plus software and open the cell_counts.cit template under Wisecaver/TEMPLATES directory. Count approximately 5000 'cells with chlorophyll' which should correspond to live *P. parvum* cells with active chloroplasts.
- 1.4 Get a post count sample weight on the analytical balance.
- 1.5 Calculate cell density (in cells per mL) with the following formula:
$$\text{Concentration} = \text{Counts} / (\text{Weight1} - \text{Weight2}) * 2$$
- 1.6 Perform a sip clean on the flow cytometer.
- 2 Place  100 μL of your algal sample in a 96-well glass bottom plate and check on the culture quality using the 60x objective. Take note of *P. parvum* cell uniformity and amount of bacterial and/or picoeukaryote contamination.
- 3 Set the ThermoMixer to  35 $^{\circ}\text{C}$
- 4 Get liquid nitrogen if starting from live cells.
- 5 Wipe down your work area with 70% ethanol:
 - Chemical fume hood
 - Benchtop
 - Rotors on the microcentrifuge and bench top spinner
 - Pipettes



Pellet Cells

35m

- 6 Collect  50 mL liquid culture (approximately 10-100 million cells) growing exponentially in a centrifuge tube and centrifuge for  4000 x g, 00:10:00 .
- 7 Pour off the supernatant into a *P. pavum* waste container, and flash freeze the cell pellet in liquid nitrogen.
- 8 Immediately add the DNazol (see step 9) **or** store in  -80 °C freezer.

It's important to work quickly here as the liquid nitrogen will have lysed many cells already. Material will be highly susceptible to environmental nucleases if allowed to sit at room temperature.

Homogenization and DNA extraction

45m

- 9 Warm the frozen centrifuge tube by cupping the area around the pellet in the palm of your hand. As soon as you see the edges of the pellet beginning to melt, add  300 µL DNazol. Pipet up and down to dislodge pellet, and transfer to a 1.5 mL centrifuge tube.
- 10 Add  5 µL RNase A. Make sure the cap is closed securely, and mix well by vigorously shaking the tube for  00:00:15 .
- 11 Incubate on the ThermoMixer at  500 rpm, 35°C, 00:20:00 . About half way through the incubation, take the tubes out and again mix well by vigorously shaking the tube for  00:00:15 . Place back in the ThermoMixer to complete the incubation step.
- 11.1 During incubation, prepare an aliquot of 100% ethanol. Prepare  300 µL per sample.
- 11.2 During incubation, prepare an ethanol and DNazol wash by mixing  1 mL DNazol with  750 µL ethanol.
- 11.3 During incubation, prepare a 75% ethanol wash using nuclease free water. Prepare  300 µL per sample.



- 12 Add  300 μL chloroform to each sample. Make sure the cap is closed securely, and mix well by vigorously shaking the tube for  00:00:15 .

15s



All liquid chloroform liquid waste should be ejected or decanted into the amber bottle labeled 'Phenol:Chloroform Waste' inside the fume hood. Any plasticware that comes into contact with chloroform needs to be properly disposed of in the sharps bin inside the fume hood labeled 'Mutagen/Carcinogen Sharps Do Not Autoclave'.

- 13 Incubate on the Hula Mixer for  00:05:00

5m

- 14 Centrifuge at  12000 rpm, 00:10:00 . Don't jostle samples when removing from the centrifuge.

10m

- 15 Transfer approximately  300 μL of the top aqueous phase to a fresh 1.5 mL tube. Pipet slowly from the top, being careful not to disrupt the interphase.



DNA Precipitation

35m

- 16 Add  225 μL of 100% ethanol. Invert ten times to mix.

- 17 Incubate at room temperature for  00:05:00 .

5m

- 18 Centrifuge at  5000 rpm, 00:04:00 . Orient the centrifuge tube so that cap hinge is on the outer edge of the rotor.

4m

Proper orientation of the tube helps to avoid the pelleted RNA when removing the supernatant in subsequent steps. Even if you can't see the pellet, you will know that it is located on the on the bottom corner under the hinge.

- 19 Discard the supernatant by taking off the initial volume with a 1 mL pipette tip. Use a 200 mL pipette tip to get the last bit of liquid.



This is the last supernatant and plastic waste that needs to be disposed of in the phenol:chloroform waste. All waste in the remaining steps can go in the standard lab 'look-alike' waste bin.



- 20 Add  300 μ L of previously prepared DNAzol wash and drag along the top of a tube rack to agitate the sample and dislodge the pellet.
- 21 Incubate at room temperature for  00:05:00 .
- 22 Centrifuge at  5000 rpm, 00:04:00 . Orient the centrifuge tube so that cap hinge is on the outer edge of the rotor.
- 23 Discard the supernatant by taking off the initial volume with a 1 mL pipette tip. Use a 200 mL pipette tip to get the last bit of liquid.
- Be sure to avoid the hinge side of the tube where the sample is.*
- 24 Add  300 μ L 75% ethanol wash and vortex briefly. If the pellet does not dislodge at this step, that is okay.
- 25 Centrifuge at  5000 rpm, 00:04:00 . Orient the centrifuge tube so that cap hinge is on the outer edge of the rotor.
- 26 Discard the supernatant by taking off the initial volume with a 1 mL pipette tip. Briefly spin down the tube using the little bench top spinner, and then use a 200 mL pipet tip to get the last bit of liquid.
- Again, even if you can't see the pellet, you will know that it is located on the on the bottom corner under the hinge. So be careful to avoid that side of the tube. Pipette away the supernatant by tipping the tube slightly towards you and pipetting the liquid from the bottom corner opposite the hinge.*
- 27 Air dry the DNA pellet for  00:05:00 by placing the tube on its side inside a folded kim wipe.
- Do not over dry the DNA pellet as this will greatly decrease its solubility. However, also be sure that all the ethanol is removed as residual ethanol will also decrease solubility.*
- 28 Resuspend sample in  50 μ L TE Buffer.
- 29 Incubate on the ThermoMixer  500 rpm, 35°C, 00:05:00 .

5m

4m

4m

5m



- 30 Briefly spin down the tube using the little bench top spinner.

QC and clean up

35m

- 31 Make sure all chloroform waste is properly disposed of in the chemical fume hood. Wipe down the chemical fume hood and the benchtop with 70-75% ethanol.

- 32 Assess DNA integrity using the TapeStation. Make sure you reserve time on the TapeStation Teams calendar.

- 33 While the TapeStation is running, measure the DNA concentration using the Qubit.

- 34 Store your sample based on on the QC results (see below).

- 34.1 If the TapeStation reports a low DNA integrity (DIN) score (< 7) and/or the DNA shows signs of banding, then the sample is too degraded to send for sequencing.

If the DIN score is low, but there is no sign of banding, the sample could still be useful in the future (e.g. for PCR). Transfer the sample to a cyrovial, label with the date and the sample identifier, and place in the 'P.parvum DNA' box in the -80 freezer. Note the location of the sample in the -80 log book.

If there is sign of banding, discard the sample.

Start a new bulk culture and try again!

- 34.2 If the concentration is less than 20 ng/uL, then there isn't sufficient quantity for novogene whole genome sequencing. However, this sample could still be useful in the future (e.g. for PCR). Transfer the sample to a cyrovial, label with the date and the sample identifier, and place in the 'P.parvum DNA' box in the -80 freezer. Note the location of the sample in the -80 log book.

Start a new bulk culture and try again!

- 34.3 If the TapeStation reports a high DNA integrity (DIN) score (≥ 7) and DNA concentration ≥ 20 ng/uL, then you have successfully isolated DNA for whole genome sequencing!!

Cap your sample tube tightly. Make sure each tube is labeled on the cap and vertically on the side with the sample's unique identifier*. Wrap the cap with parafilm and place sample in the 'Novogene' box in the full sized -20 freezer.

Update the purple columns in the 'Prymnesium strain tracking' google sheet.

**The identifier should start with a letter and be no more than 12 characters long. Write it carefully using a fine tip permanent marker as this tube will be shipped off for DNA sequencing.*